

# SLIC Index V3

Methodology Technical Paper

English master edition

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This paper documents the live published SLIC scorer as implemented in **scripts/rescore-all-cities.mjs** and written to **src/data/publishedRankingData.json**. It replaces earlier draft descriptions of a 35-indicator anchor model that no longer matches the public dataset.

| Snapshot fact                   | Value |
|---------------------------------|-------|
| Cities in public file           | 163   |
| Ranked cities                   | 154   |
| Watchlist cities                | 9     |
| Scored metric lines             | 20    |
| Visible non-scoring diagnostics | 3     |

# 1. Published method

SLIC has five public pillars. Within each pillar, observed scored metrics are combined by weighted mean. Across pillars, the final score uses a weighted AMPI, so cross-pillar imbalance is punished mathematically rather than rhetorically.

```
normalize(x, p05, p95, positive) = 100 * clamp((winsor(x) - p05) / (p95 - p05), 0, 1)
normalize(x, p05, p95, negative) = 100 * clamp((p95 - winsor(x)) / (p95 - p05), 0, 1)

P_p(c) = sum observed alpha_(p,m) * q_m(c) / sum observed alpha_(p,m)

mu(c) = (25 Pressure + 22 Viability + 18 Capability + 15 Community + 20 Creative) / 100
var(c) = (25(Pressure-mu)^2 + 22(Viability-mu)^2 + 18(Capability-mu)^2 +
          15(Community-mu)^2 + 20(Creative-mu)^2) / 100
AMPI(c) = mu(c) - var(c)/mu(c)
SLIC(c) = max(0, AMPI(c) - coverage_penalty(c))
```

The final published score is therefore **not** a simple weighted average of the displayed pillar scores. That distinction matters: a city with one catastrophic pillar is supposed to score worse than a city with the same weighted mean but more even performance.

# 2. What is scored

The current public model scores **20** metric lines and keeps **3** additional metric lines visible as diagnostics only. Visibility and scoring are intentionally separated.

## Growth (25)

| Metric                             | Weight | Status          | Method note                                                                                 |
|------------------------------------|--------|-----------------|---------------------------------------------------------------------------------------------|
| Tax-adjusted PPP disposable income | 9      | Scored          | Residual money after essentials, converted once through the PPP private-consumption factor. |
| Housing burden                     | 5      | Scored          | Housing cost share of income; higher burden scores worse.                                   |
| Household debt burden              | 4      | Scored          | Debt relative to income, with fallback when city debt evidence is thin.                     |
| Working time pressure              | 4      | Scored          | Higher work burden scores worse.                                                            |
| Suicide and severe mental strain   | 3      | Scored          | Public-health strain proxy; higher severe-strain burden scores worse.                       |
| Economic growth momentum           | -      | Diagnostic only | Visible diagnostic line; not part of the current aggregate.                                 |

## Viability (22)

| Metric                                     | Weight | Status          | Method note                                                 |
|--------------------------------------------|--------|-----------------|-------------------------------------------------------------|
| Personal safety                            | 5      | Scored          | Harm and victimization outcomes, not surveillance theatre.  |
| Transit access and commute burden          | 5      | Scored          | Better access and lower burden score better.                |
| Clean air                                  | 4      | Scored          | Higher pollution exposure scores worse.                     |
| Water, sanitation, and utility reliability | 4      | Scored          | Safer access and higher reliability score better.           |
| Digital infrastructure                     | 4      | Scored          | Broadband quality and affordability.                        |
| Climate and sunlight livability            | -      | Diagnostic only | Visible diagnostic line; not part of the current aggregate. |

## Capability (18)

| Metric             | Weight | Status | Method note                            |
|--------------------|--------|--------|----------------------------------------|
| Healthcare quality | 8      | Scored | Effective-care quality and access.     |
| Education quality  | 6      | Scored | Learning quality and skills formation. |

| Metric                                        | Weight | Status | Method note                                                                         |
|-----------------------------------------------|--------|--------|-------------------------------------------------------------------------------------|
| Equal opportunity and distributional fairness | 4      | Scored | Composite scored metric: 70% equal-opportunity evidence, 30% reversed Gini context. |

## Community (15)

| Metric                                       | Weight | Status          | Method note                                                                                                 |
|----------------------------------------------|--------|-----------------|-------------------------------------------------------------------------------------------------------------|
| Hospitality and belonging                    | 5      | Scored          | Do people feel welcome in practice?                                                                         |
| Tolerance and pluralism                      | 5      | Scored          | Composite scored metric: 40% Equaldex, 30% Freedom House, 30% reversed hate-crime or civil-liberties proxy. |
| Cultural, historic, and public-life vitality | 5      | Scored          | Everyday public-life texture; visitor demand is contextual, not an automatic bonus.                         |
| Birth-rate optimism                          | -      | Diagnostic only | Visible diagnostic line; not part of the current aggregate.                                                 |

## Creative (20)

| Metric                                   | Weight | Status | Method note                                                                             |
|------------------------------------------|--------|--------|-----------------------------------------------------------------------------------------|
| Entrepreneurial dynamism                 | 6      | Scored | Startup formation and productive entry.                                                 |
| Innovation and research intensity        | 5      | Scored | Research depth and knowledge production.                                                |
| Economic vitality and productive context | 5      | Scored | Composite scored metric: 50% investment signal, 30% GDP per capita PPP, 20% GDP growth. |
| Administrative and investment friction   | 4      | Scored | Higher time-cost and licensing burden score worse.                                      |

### 3. Composite metrics

Three scored terms are composites. When one component is missing, the composite is reweighted over the observed component weights only; nothing is silently imputed.

```
EqualOpportunityFairness = 0.7 * EqualOpportunity + 0.3 * ReverseGini
TolerancePluralism      = 0.4 * Equaldex + 0.3 * FreedomHouse + 0.3 * ReverseHateCrime
EconomicVitality        = 0.5 * InvestmentSignal + 0.3 * GDPperCapitaPPP + 0.2 * GDPGrowth
```

### 4. Coverage, watchlist logic, and ranking protocol

```
cov_direct = 1 if observed else 0
cov_composite = observed component weight / total component weight

Cov_p(c) = sum alpha_(p,m) * cov_m(c) / sum alpha_(p,m)
Cov(c)    = (25 Cov_pressure + 22 Cov_viability + 18 Cov_capability +
             15 Cov_community + 20 Cov_creative) / 100
```

| Coverage grade | Rule                                    | Penalty | Rank status |
|----------------|-----------------------------------------|---------|-------------|
| A              | Cov >= 0.75                             | 0       | Ranked      |
| B              | 0.50 <= Cov < 0.75                      | 5       | Ranked      |
| C              | 0.35 <= Cov < 0.50                      | 15      | Ranked      |
| Watchlist      | Cov < 0.35 or manual watchlist override | 0       | Watchlist   |

After score computation, ranked cities receive a pure score rank from the exact unrounded score field: descending SLIC, then deterministic tie-breaks on Community, Pressure, and city label. Alpha, Beta, and Gamma are then computed as a separate public overlay. The overlay allows one city per country across all three tiers, except Taiwan may hold 2 public-tier seats; Alpha requires Community >= 40 and Pressure >= 40, with Europe capped at 2 Alpha seats, Oceania capped at 1, South Korea capped at 1, Japan capped at 0, and Israel excluded from Alpha; Beta raises the floors to Community >= 45 and Pressure >= 45; Gamma fills from the remaining ranked cities.

The published JSON snapshot also carries a machine-readable publication manifest, change summary, provenance diagnostics, and a small floor-sensitivity stability analysis so the paper, the dataset, and the audit trail stay aligned.

Live example: Singapore is currently pure rank **#19** with SLIC **60.2**, but its Community (38.8) and Pressure (43.3) place it in **Gamma**. Bangkok is pure rank **#27** with SLIC **58.5**, yet enters **Alpha** because Community (90.0) and Pressure (45.4) both clear the Alpha floor.

### 5. Worked example from the live dataset

Example city: **Kaohsiung**. This walkthrough uses the published row in the live dataset, not a fictional demonstration.

```
Negative-direction normalization example (personal safety)
raw = 0.8
p05 = 0.23
p95 = 22.63
score = 100 * (22.63 - 0.8) / (22.63 - 0.23)
score = 97.5

Growth weighted mean
Growth = (9x81.5 + 5x81.1 + 4x41.4 + 3x27.5) / 21
Growth = 66.1

Cross-pillar AMPI
mu = 70.307
var = 252.168
AMPI = 70.307 - 252.168 / 70.307
AMPI = 66.720
Coverage = 0.77 -> grade A -> penalty 0
Published SLIC = 66.7
```

The important methodological point is visible in the arithmetic: the weighted pillar mean for this city is higher than the final score because AMPI makes uneven pillar structure costly.

## 6. Data profile of the current public file

| Measure                               | Value |
|---------------------------------------|-------|
| City / metro metric share             | 48.8% |
| National / international metric share | 26.0% |
| Composite metric share                | 15.7% |
| Derived metric share                  | 9.6%  |
| Grade A cities                        | 138   |
| Grade B cities                        | 2     |
| Grade C cities                        | 16    |
| Watchlist cities                      | 9     |

These shares are computed directly from non-null metric lines in the published JSON. They are included here so the paper does not rely on invented methodology graphics or unsupported source-mix claims.

## 7. Published ranked cities

The table below is the ranked slice of the current published dataset. Watchlist cities are excluded from this list by definition.

| #  | City        | Country        | SLIC | G    | V    | Cap  | Com  | Cr   | Grade |
|----|-------------|----------------|------|------|------|------|------|------|-------|
| 2  | Raleigh     | United States  | 69.0 | 75.1 | 76.0 | 79.1 | 53.8 | 62.0 | A     |
| 3  | Montreal    | Canada         | 68.9 | 62.3 | 93.3 | 75.3 | 74.1 | 54.1 | A     |
| 4  | Kaohsiung   | Taiwan         | 66.7 | 66.1 | 86.5 | 90.6 | 47.1 | 56.9 | A     |
| 5  | Minneapolis | United States  | 64.6 | 71.6 | 69.1 | 79.1 | 44.2 | 62.0 | A     |
| 6  | Pittsburgh  | United States  | 64.5 | 76.0 | 65.1 | 79.1 | 50.2 | 55.7 | A     |
| 7  | Eindhoven   | Netherlands    | 64.3 | 64.6 | 97.6 | 71.8 | 65.7 | 42.8 | A     |
| 8  | Graz        | Austria        | 63.9 | 62.4 | 87.1 | 74.4 | 73.4 | 41.4 | A     |
| 9  | Taipei      | Taiwan         | 63.8 | 48.4 | 89.5 | 89.7 | 47.1 | 69.4 | A     |
| 10 | Prague      | Czechia        | 63.5 | 62.0 | 90.6 | 61.9 | 87.7 | 42.4 | A     |
| 11 | Ottawa      | Canada         | 63.5 | 55.3 | 93.3 | 75.3 | 75.6 | 44.2 | A     |
| 12 | Jeju City   | South Korea    | 61.6 | 56.2 | 99.2 | 85.8 | 44.8 | 50.6 | A     |
| 13 | Braga       | Portugal       | 60.6 | 58.4 | 95.7 | 68.8 | 57.9 | 43.8 | A     |
| 14 | Lyon        | France         | 60.6 | 52.4 | 97.9 | 63.9 | 73.1 | 45.1 | A     |
| 15 | Incheon     | South Korea    | 60.5 | 53.3 | 99.2 | 85.8 | 44.8 | 50.6 | A     |
| 16 | Milan       | Italy          | 60.5 | 88.5 | 85.8 | 66.9 | 58.4 | 29.4 | A     |
| 17 | Busan       | South Korea    | 60.5 | 53.2 | 99.2 | 85.8 | 44.8 | 50.6 | A     |
| 18 | Suwon       | South Korea    | 60.3 | 53.3 | 93.1 | 85.8 | 45.1 | 50.6 | A     |
| 19 | Singapore   | Singapore      | 60.2 | 43.3 | 87.4 | 94.1 | 38.8 | 73.3 | A     |
| 20 | Copenhagen  | Denmark        | 59.9 | 33.1 | 97.9 | 71.3 | 89.6 | 61.5 | A     |
| 21 | Perth       | Australia      | 59.8 | 45.3 | 89.5 | 87.1 | 56.3 | 51.2 | A     |
| 22 | Tokyo       | Japan          | 59.5 | 46.2 | 94.2 | 76.8 | 70.2 | 44.2 | A     |
| 23 | Toronto     | Canada         | 59.3 | 38.7 | 93.3 | 75.3 | 72.5 | 54.1 | A     |
| 24 | Valparaiso  | Chile          | 58.9 | 58.3 | 72.7 | 73.6 | 56.9 | 43.0 | A     |
| 25 | Porto       | Portugal       | 58.6 | 52.5 | 95.7 | 68.8 | 57.9 | 43.8 | A     |
| 26 | Chicago     | United States  | 58.6 | 56.0 | 56.6 | 79.1 | 47.6 | 62.0 | A     |
| 27 | Bangkok     | Thailand       | 58.5 | 45.4 | 84.9 | 62.5 | 90.0 | 46.0 | A     |
| 28 | Adelaide    | Australia      | 58.5 | 43.7 | 89.5 | 87.1 | 56.3 | 49.2 | A     |
| 29 | Zurich      | Switzerland    | 58.3 | 39.1 | 97.6 | 72.7 | 76.0 | 49.4 | A     |
| 30 | London      | United Kingdom | 58.2 | 35.1 | 82.8 | 78.4 | 70.2 | 59.3 | A     |
| 31 | Melbourne   | Australia      | 58.2 | 36.4 | 89.5 | 87.1 | 56.3 | 60.5 | A     |
| 32 | Brisbane    | Australia      | 58.2 | 39.7 | 89.5 | 87.1 | 56.3 | 54.4 | A     |
| 33 | Tallinn     | Estonia        | 58.2 | 51.0 | 90.1 | 58.4 | 49.0 | 57.8 | A     |
| 34 | Helsinki    | Finland        | 58.2 | 40.9 | 91.5 | 81.0 | 57.5 | 52.9 | A     |
| 35 | Vienna      | Austria        | 58.1 | 46.8 | 87.1 | 74.4 | 71.1 | 41.4 | A     |
| 37 | Santiago    | Chile          | 57.9 | 50.7 | 72.7 | 83.8 | 56.9 | 44.6 | A     |
| 38 | New York    | United States  | 57.7 | 27.1 | 85.4 | 82.0 | 68.5 | 75.3 | A     |

| #  | City          | Country              | SLIC | G    | V    | Cap  | Com  | Cr   | Grade |
|----|---------------|----------------------|------|------|------|------|------|------|-------|
| 39 | Fukuoka       | Japan                | 57.6 | 59.0 | 94.5 | 67.4 | 46.6 | 42.3 | A     |
| 40 | Dunedin       | New Zealand          | 57.5 | 51.2 | 92.6 | 85.7 | 57.2 | 37.1 | A     |
| 41 | Christchurch  | New Zealand          | 57.2 | 45.8 | 92.6 | 85.7 | 57.2 | 41.9 | A     |
| 42 | Vancouver     | Canada               | 57.1 | 34.7 | 93.3 | 75.3 | 70.2 | 54.1 | A     |
| 43 | Venice        | Italy                | 56.8 | 53.9 | 84.7 | 67.8 | 68.6 | 34.8 | A     |
| 44 | Torun         | Poland               | 56.0 | 68.4 | 79.5 | 65.1 | 52.0 | 31.9 | A     |
| 45 | Abu Dhabi     | United Arab Emirates | 55.9 | 70.5 | 92.6 | 62.0 | 29.5 | 46.2 | A     |
| 46 | Wellington    | New Zealand          | 55.9 | 37.8 | 92.6 | 85.7 | 57.2 | 48.8 | A     |
| 47 | Dubai         | United Arab Emirates | 55.8 | 69.8 | 92.6 | 62.0 | 29.5 | 46.2 | A     |
| 49 | Hiroshima     | Japan                | 55.8 | 59.0 | 94.5 | 67.4 | 46.6 | 37.3 | A     |
| 50 | Sapporo       | Japan                | 55.8 | 59.0 | 94.5 | 67.4 | 46.6 | 37.3 | A     |
| 51 | Katowice      | Poland               | 55.4 | 65.9 | 79.5 | 65.1 | 52.0 | 31.9 | A     |
| 52 | Kobe          | Japan                | 55.4 | 58.0 | 94.5 | 67.4 | 46.6 | 37.3 | A     |
| 53 | Manama        | Bahrain              | 55.3 | 66.4 | 79.7 | 57.8 | 39.7 | 42.5 | A     |
| 54 | Sydney        | Australia            | 54.5 | 27.8 | 89.5 | 87.1 | 56.3 | 63.9 | A     |
| 55 | Auckland      | New Zealand          | 54.4 | 32.4 | 92.6 | 85.7 | 57.2 | 52.9 | A     |
| 56 | Gdansk        | Poland               | 54.3 | 60.9 | 79.5 | 65.1 | 52.0 | 31.9 | A     |
| 57 | Bologna       | Italy                | 54.0 | 56.4 | 88.2 | 66.9 | 58.4 | 29.4 | A     |
| 58 | Amsterdam     | Netherlands          | 53.9 | 37.6 | 97.6 | 71.8 | 65.7 | 42.8 | A     |
| 59 | Bratislava    | Slovakia             | 53.9 | 57.3 | 89.7 | 49.7 | 63.2 | 35.3 | A     |
| 60 | Hobart        | Australia            | 52.5 | 33.8 | 89.5 | 87.1 | 56.3 | 45.1 | A     |
| 61 | Krakow        | Poland               | 52.3 | 53.5 | 79.5 | 65.1 | 52.0 | 31.9 | A     |
| 62 | Port Louis    | Mauritius            | 51.2 | 56.0 | 76.2 | 47.0 | 39.6 | 45.4 | A     |
| 63 | Melaka        | Malaysia             | 49.5 | 63.0 | 75.3 | 52.4 | 40.3 | 31.5 | A     |
| 64 | Bucharest     | Romania              | 49.5 | 57.5 | 81.4 | 50.6 | 40.7 | 34.8 | A     |
| 65 | Kuching       | Malaysia             | 49.3 | 61.5 | 73.7 | 52.4 | 41.1 | 31.5 | A     |
| 66 | Kota Kinabalu | Malaysia             | 49.3 | 61.5 | 77.9 | 52.4 | 40.0 | 31.5 | A     |
| 67 | Montevideo    | Uruguay              | 49.1 | 52.4 | 59.5 | 74.6 | 64.6 | 24.4 | A     |
| 68 | Khobar        | Saudi Arabia         | 48.9 | 91.3 | 96.3 | 78.9 | 17.2 | 24.1 | A     |
| 69 | Jeddah        | Saudi Arabia         | 48.8 | 90.6 | 96.3 | 78.9 | 17.2 | 24.1 | A     |
| 70 | Riyadh        | Saudi Arabia         | 48.7 | 89.9 | 96.3 | 78.9 | 17.2 | 24.1 | A     |
| 71 | Paris         | France               | 48.6 | 27.7 | 97.9 | 63.9 | 69.2 | 45.1 | A     |
| 72 | George Town   | Malaysia             | 48.6 | 58.0 | 73.7 | 52.4 | 41.1 | 31.5 | A     |
| 73 | San Juan      | Puerto Rico          | 48.2 | 43.8 | 59.2 | 68.3 | 63.3 | 45.8 | B     |
| 74 | Budapest      | Hungary              | 48.1 | 47.2 | 93.1 | 55.5 | 63.5 | 26.2 | A     |
| 75 | Kuala Lumpur  | Malaysia             | 46.8 | 51.0 | 73.7 | 52.4 | 41.1 | 31.5 | A     |
| 76 | Hat Yai       | Thailand             | 46.6 | 55.1 | 66.7 | 56.4 | 52.2 | 22.8 | A     |
| 77 | Salalah       | Oman                 | 45.4 | 73.9 | 71.1 | 42.0 | 31.2 | 30.6 | A     |

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|-----|-----------------|-------------|------|------|------|------|------|------|-------|
| 78  | San Jose        | Costa Rica  | 45.3 | 45.7 | 43.5 | 66.0 | 47.9 | 35.9 | A     |
| 79  | Male            | Maldives    | 45.0 | 54.6 | 72.1 | 39.8 | 35.8 | 35.4 | A     |
| 80  | Phuket          | Thailand    | 44.9 | 48.0 | 68.0 | 56.4 | 52.2 | 22.8 | A     |
| 81  | Doha            | Qatar       | 44.7 | 59.0 | 73.9 | 54.4 | 18.6 | 37.6 | A     |
| 82  | Muscat          | Oman        | 44.7 | 66.9 | 71.1 | 42.0 | 31.2 | 30.6 | A     |
| 83  | Cordoba         | Argentina   | 44.5 | 67.5 | 69.2 | 78.6 | 46.8 | 10.2 | A     |
| 84  | Panama City     | Panama      | 44.2 | 61.0 | 44.0 | 64.2 | 41.8 | 26.9 | A     |
| 85  | Chiang Mai      | Thailand    | 43.8 | 54.3 | 49.7 | 56.4 | 52.2 | 22.8 | A     |
| 86  | Belgrade        | Serbia      | 43.1 | 46.5 | 74.8 | 53.2 | 32.1 | 30.3 | A     |
| 87  | Tianjin         | China       | 42.4 | 43.5 | 98.6 | 71.9 | 19.7 | 37.1 | A     |
| 88  | Curitiba        | Brazil      | 42.2 | 51.8 | 45.1 | 69.5 | 36.3 | 27.6 | A     |
| 89  | Kuwait City     | Kuwait      | 42.2 | 91.4 | 69.2 | 65.0 | 19.7 | 19.9 | A     |
| 90  | Florianopolis   | Brazil      | 41.9 | 50.4 | 45.1 | 69.5 | 36.3 | 27.6 | A     |
| 91  | Buenos Aires    | Argentina   | 41.8 | 55.4 | 69.2 | 78.6 | 46.8 | 10.2 | A     |
| 92  | Chongqing       | China       | 41.0 | 38.5 | 75.2 | 71.9 | 20.3 | 37.1 | A     |
| 93  | Chengdu         | China       | 40.9 | 37.8 | 77.0 | 71.9 | 20.5 | 37.1 | A     |
| 94  | Guangzhou       | China       | 40.0 | 38.1 | 98.6 | 71.9 | 19.7 | 37.1 | A     |
| 95  | Sao Paulo       | Brazil      | 39.9 | 42.7 | 45.1 | 69.5 | 36.3 | 27.6 | A     |
| 96  | Hangzhou        | China       | 39.0 | 33.5 | 84.0 | 71.9 | 20.6 | 37.1 | A     |
| 97  | Shenzhen        | China       | 38.7 | 35.3 | 98.6 | 71.9 | 19.7 | 37.1 | A     |
| 98  | Nanjing         | China       | 38.4 | 52.6 | 75.3 | 59.0 | 7.1  | 34.5 | A     |
| 100 | Vilnius         | Lithuania   | 36.9 | 65.0 | -    | 41.6 | 74.0 | 41.4 | C     |
| 101 | Shanghai        | China       | 36.7 | 31.3 | 98.6 | 71.9 | 19.7 | 37.1 | A     |
| 102 | Arequipa        | Peru        | 36.3 | 57.5 | 34.1 | 42.3 | 34.9 | 24.7 | A     |
| 103 | Lima            | Peru        | 35.8 | 51.3 | 34.1 | 42.3 | 34.9 | 24.7 | A     |
| 104 | Gaborone        | Botswana    | 35.7 | 51.4 | 31.9 | 22.2 | 45.2 | 39.0 | A     |
| 105 | Nizhny Novgorod | Russia      | 35.7 | 70.7 | 62.4 | 59.2 | 15.9 | 15.5 | A     |
| 106 | Suva            | Fiji        | 35.7 | 45.8 | 36.1 | 47.7 | 30.8 | 25.1 | A     |
| 107 | Medellin        | Colombia    | 35.0 | 54.4 | 28.0 | 66.3 | 41.9 | 19.6 | A     |
| 108 | Bogota          | Colombia    | 34.1 | 48.7 | 28.0 | 66.3 | 41.9 | 19.6 | A     |
| 109 | Cebu City       | Philippines | 33.2 | 55.8 | 38.2 | 40.4 | 30.8 | 17.5 | A     |
| 110 | Makati          | Philippines | 33.0 | 52.1 | 38.2 | 40.4 | 30.8 | 17.5 | A     |
| 111 | Windhoek        | Namibia     | 32.9 | 42.7 | 30.5 | 32.1 | 39.3 | 25.4 | A     |
| 112 | Jakarta         | Indonesia   | 32.7 | 58.6 | 40.8 | 35.1 | 29.9 | 17.8 | A     |
| 113 | Surabaya        | Indonesia   | 32.7 | 63.7 | 40.8 | 35.1 | 29.9 | 17.8 | A     |
| 114 | Rabat           | Morocco     | 32.4 | 46.9 | 59.9 | 34.1 | 24.2 | 18.8 | A     |
| 115 | Riga            | Latvia      | 32.4 | 60.6 | -    | 33.5 | 76.2 | 40.3 | C     |
| 116 | Casablanca      | Morocco     | 32.3 | 46.0 | 59.9 | 34.1 | 24.2 | 18.8 | A     |
| 117 | Göteborg        | Sweden      | 31.5 | 65.3 | -    | 14.5 | 95.5 | 63.3 | C     |
| 118 | Hyderabad       | India       | 31.3 | 44.7 | 53.1 | 30.4 | 33.5 | 14.6 | A     |



| #   | City         | Country      | SLIC | G    | V    | Cap  | Com  | Cr   | Grade |
|-----|--------------|--------------|------|------|------|------|------|------|-------|
| 119 | Pune         | India        | 31.2 | 44.4 | 53.1 | 30.4 | 33.5 | 14.6 | A     |
| 120 | Bengaluru    | India        | 31.1 | 43.9 | 53.1 | 30.4 | 33.5 | 14.6 | A     |
| 122 | Chandigarh   | India        | 30.6 | 43.9 | 44.1 | 30.4 | 34.1 | 14.6 | A     |
| 123 | Moscow       | Russia       | 30.5 | 41.0 | 62.4 | 59.2 | 15.9 | 15.5 | A     |
| 124 | Kigali       | Rwanda       | 30.5 | 56.0 | 47.2 | 15.8 | 30.5 | 24.8 | A     |
| 125 | Kandy        | Sri Lanka    | 29.0 | 51.5 | 40.7 | 33.7 | 28.6 | 11.7 | A     |
| 126 | Aqaba        | Jordan       | 28.9 | 43.2 | 66.0 | 46.5 | 11.5 | 17.0 | A     |
| 127 | Colombo      | Sri Lanka    | 28.9 | 48.8 | 40.7 | 33.7 | 28.6 | 11.7 | A     |
| 128 | Thimphu      | Bhutan       | 28.6 | 47.2 | 47.0 | 25.2 | 40.4 | 10.2 | A     |
| 129 | Da Nang      | Vietnam      | 28.2 | 51.7 | -    | 44.3 | 28.1 | 20.8 | B     |
| 130 | Amman        | Jordan       | 28.1 | 40.3 | 66.0 | 46.5 | 11.5 | 17.0 | A     |
| 131 | Tbilisi      | Georgia      | 26.7 | 50.0 | -    | 33.1 | 40.9 | 43.1 | C     |
| 132 | Valencia     | Spain        | 26.6 | 59.8 | -    | 31.1 | 87.1 | 30.4 | C     |
| 133 | Johannesburg | South Africa | 26.3 | 34.3 | 15.1 | 26.3 | 42.3 | 29.8 | A     |
| 134 | Cape Town    | South Africa | 26.2 | 34.2 | 15.1 | 26.3 | 42.3 | 29.8 | A     |
| 135 | Zagreb       | Croatia      | 25.7 | 61.9 | -    | 17.8 | 77.4 | 42.3 | C     |
| 136 | Accra        | Ghana        | 25.3 | 55.6 | 35.9 | 14.9 | 30.3 | 17.8 | A     |
| 137 | Kumasi       | Ghana        | 25.1 | 59.4 | 35.9 | 14.9 | 30.3 | 17.8 | A     |
| 138 | Merida       | Mexico       | 25.1 | 53.7 | 13.3 | 55.5 | 40.5 | 14.1 | A     |
| 139 | Guadalajara  | Mexico       | 24.8 | 49.9 | 13.3 | 55.5 | 40.5 | 14.1 | A     |
| 140 | Cork         | Ireland      | 24.5 | 82.1 | -    | 13.3 | 91.8 | 37.4 | C     |
| 142 | Nairobi      | Kenya        | 23.9 | 48.5 | 45.1 | 12.5 | 28.1 | 14.5 | A     |
| 143 | Mexico City  | Mexico       | 23.3 | 39.2 | 13.3 | 55.5 | 40.5 | 14.1 | A     |
| 145 | Pokhara      | Nepal        | 22.3 | 44.3 | 37.6 | 12.2 | 21.7 | 16.6 | A     |
| 146 | Kathmandu    | Nepal        | 22.3 | 42.3 | 37.6 | 12.2 | 21.7 | 16.6 | A     |
| 147 | Chattogram   | Bangladesh   | 21.7 | 51.3 | 48.3 | 10.8 | 19.6 | 15.7 | A     |
| 148 | Bergen       | Norway       | 21.7 | 76.4 | -    | 3.2  | 97.8 | 48.8 | C     |
| 149 | Dhaka        | Bangladesh   | 21.6 | 49.3 | 48.3 | 10.8 | 19.6 | 15.7 | A     |
| 151 | Asuncion     | Paraguay     | 17.5 | 40.4 | -    | 74.7 | 45.4 | 14.9 | C     |
| 152 | Antwerp      | Belgium      | 16.9 | 65.1 | -    | 4.4  | 87.9 | 40.4 | C     |
| 153 | Phnom Penh   | Cambodia     | 16.6 | 43.9 | -    | 18.2 | 34.5 | 36.3 | C     |
| 154 | Brno         | Czechia      | 15.5 | 70.5 | -    | 0.0  | 84.3 | 42.4 | C     |
| 155 | Dakar        | Senegal      | 11.4 | 40.5 | 0.7  | 10.6 | 26.3 | 26.7 | A     |
| 156 | Munich       | Germany      | 10.8 | 18.3 | -    | 27.1 | 65.0 | 36.2 | C     |
| 157 | Islamabad    | Pakistan     | 10.7 | 47.5 | 32.7 | 8.1  | 13.0 | 4.4  | A     |
| 158 | Karachi      | Pakistan     | 10.6 | 48.0 | 32.7 | 8.1  | 13.0 | 4.4  | A     |
| 159 | Lahore       | Pakistan     | 10.6 | 48.3 | 32.7 | 8.1  | 13.0 | 4.4  | A     |
| 161 | Ljubljana    | Slovenia     | 10.2 | 67.0 | -    | 0.0  | 90.6 | 33.2 | C     |
| 162 | Kampala      | Uganda       | 8.7  | 31.8 | -    | 68.7 | 15.1 | 22.5 | C     |
| 163 | Alexandria   | Egypt        | 2.4  | 45.0 | -    | 11.3 | 7.7  | 30.6 | C     |